Delta Physics

Select a topic of your Choice

5 minutes – Individual

Gravitation

Why doesn't the Space Station fall down?



Higher Physics

PART 1

Satellites in Free Fall

The Space Station is 400 km up, and gravity there is still strong. Why doesn't it fall down?

By the end of this part…



1. I can **explain** how a **satellite** stays in **orbit**, using **Newton's cannon**
2. I can **state** that a **satellite** is in **free fall**
3. I can **explain** why **astronauts** feel **weightless**

Projectiles Recap

1. **State** the **horizontal acceleration** of a projectile. (Ignore air resistance.)

   Zero: constant horizontal velocity.

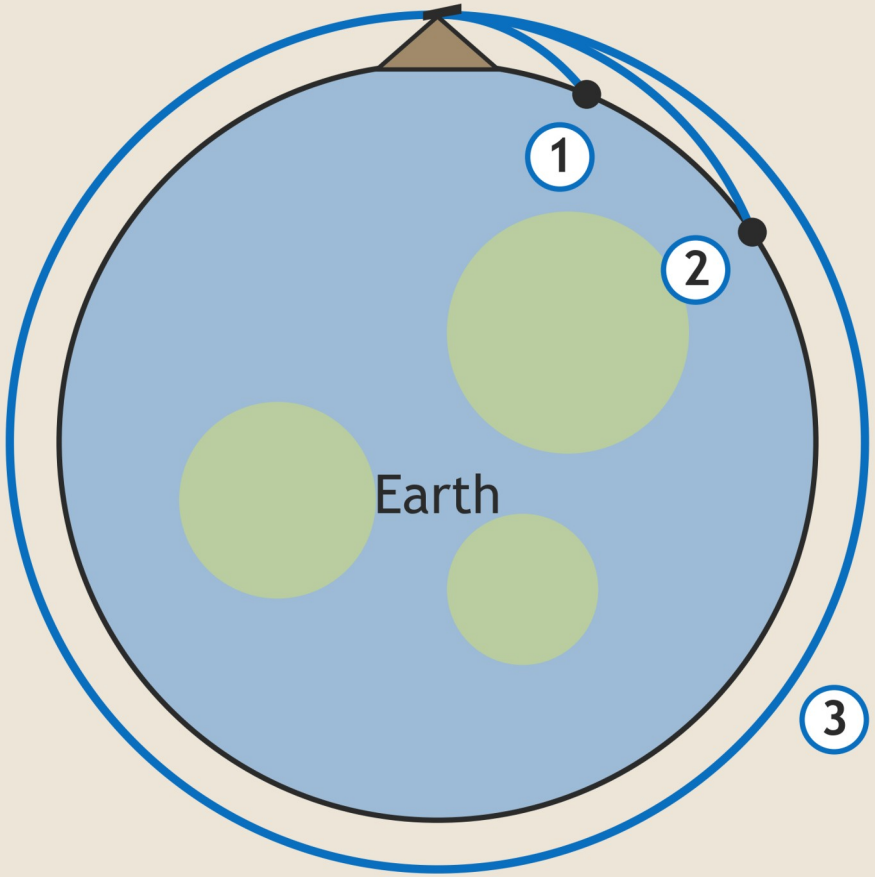
2. **State** the **vertical acceleration** of a projectile near the Earth.

   9.8 m s^{-2}, downwards.

3. One ball is **dropped** as another is **fired sideways** from the same height. **State** which lands first.

   They land at the same time.

Newton's Cannon



1 **Slow**: it lands near the mountain.

2 **Faster**: it lands further away.

3 **Fast enough**: the ground curves away as fast as it falls. It never lands: an **orbit**.

Newton's Cannon in 1728

V

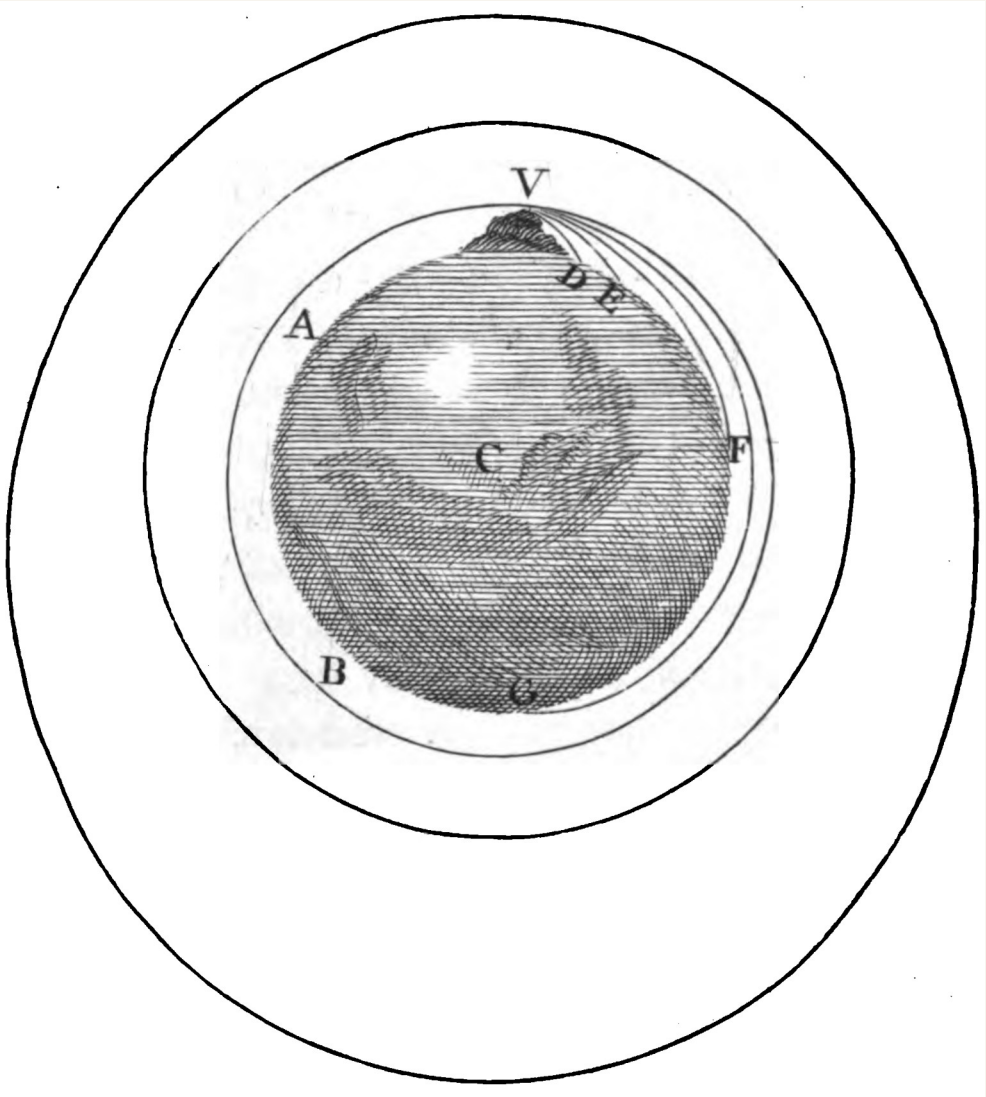
the mountain top

D, E, F, B

landing points, faster each time

Outer circles

orbits: the ball never lands



Satellites Are in Free Fall

Free fall

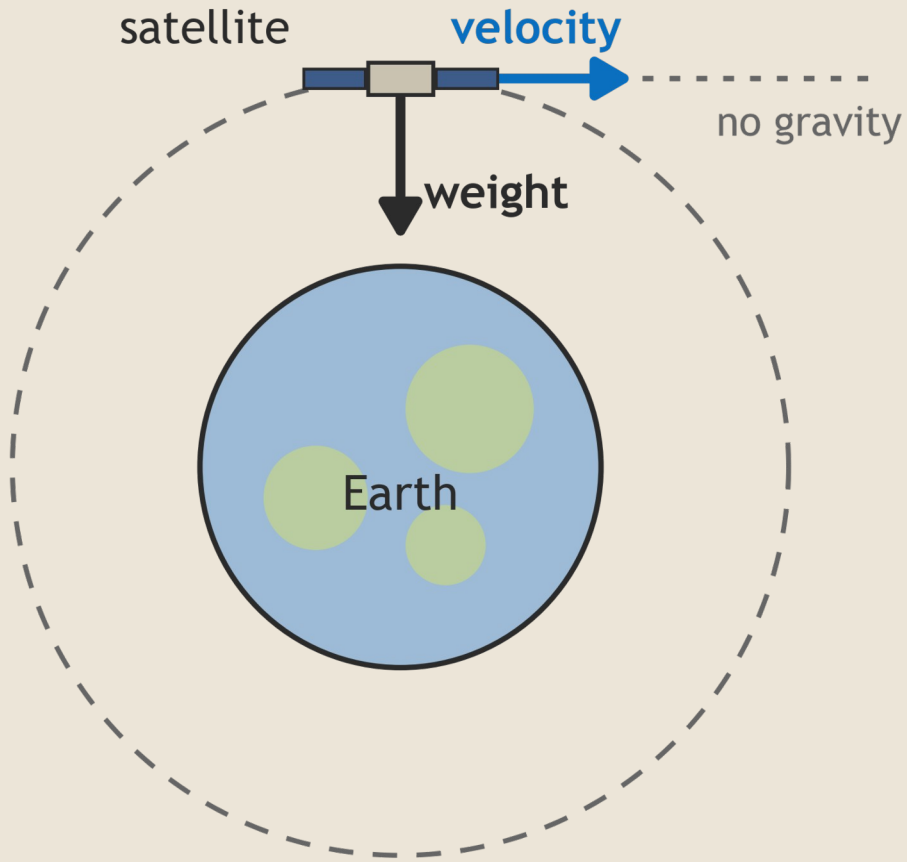
weight is the **only force** acting

A satellite

is in **free fall** around a **planet** or **star**

It stays up

it moves round as it **falls**

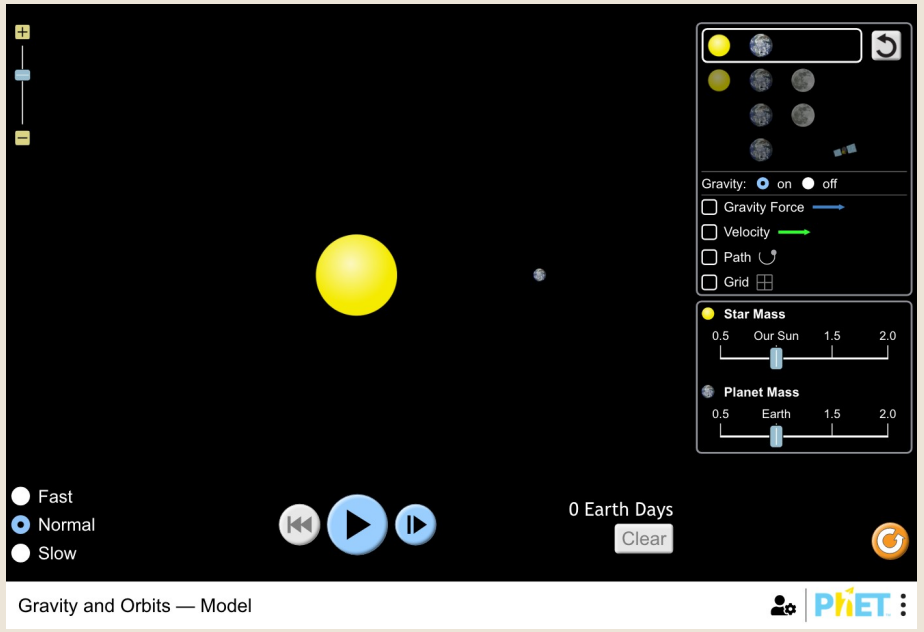


Simulation: Gravity and Orbits

Open the Model screen. Tick **Velocity** and **Path**.

Change one thing. **Describe** the path.

Change	Path
Gravity off	a straight line
Slower	smaller oval, or crashes
Faster	bigger oval, or escapes



Scan to open the sim.

Weightlessness

Not zero gravity

astronauts still have **weight**

They float

astronaut and ISS **fall together**

So

the floor exerts **no force** on them



Your Turn



1. **Explain** why the ISS does not fall to the ground.

It is falling, but the Earth curves away.

2. **State** what is meant by a satellite being in **free fall**.

Weight is the only force on it.

3. **Explain** why an astronaut floats inside the ISS.

Both fall together: the floor exerts no force.

2026 Paper 1 Q7 (page 7). The ISS is orbiting the Earth at a height of 420 km. Which of these statements is/are correct?



A I only
B I and II only
C I and III only
D II and III only
E I, II and III

I The ISS is accelerating towards the Earth.
II The magnitude of the vertical velocity of the ISS increases as it is orbiting the Earth.
III The gravitational field strength is zero at a height of 420 km above the Earth's surface.

Answer A: it always accelerates towards the Earth.

Scan: the 2026 paper, page 7.

Look back — by the end of this part...



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3. I can **explain** why **astronauts** feel **weightless**

Course specification — 1 of 1

- Knowledge that satellites are in free fall around a planet/star.

Scan: the course specification, page 8.

Credits

Title: the ISS: NASA

Newton's Cannon in 1728: Isaac Newton, A Treatise of the System of the World (1728), public domain, via Wikimedia Commons

Simulation: Gravity and Orbits: PhET Interactive Simulations, University of Colorado Boulder, https://phet.colorado.edu, CC BY-NC 4.0

Weightlessness: NASA

2026 Paper 1 Q7: © Qualifications Scotland (SQA). 2026 Higher Physics Paper 1, Q7

Course specification — 1 of 1: © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8

R Stewart - November 2026